

Chp #2

Indifference curve and Budget line

Topic:- Income and substitution
effect.



Zarrah Mushtaq

Income and Substitution effects

A price effect includes both an income effect and substitution effect.

If the price of a product falls, it means that it is now cheaper relative to other products. This means that a consumer is likely to purchase more of the product. The consumer is therefore likely to substitute purchases of products whose price has fallen for other more expensive products. This is known as substitution effect.

The fall in the price of a product will also mean that a consumer's real income has effectively risen and so that more of all products, including one whose price has fallen, can now be purchased. This is known as income effect.

Normal good, the income effect will operate so that consumption of the product whose price has fallen will increase. Income and substitution effect goes in same direction. If the product is an inferior good, the income effect will operate so the consumption of the product whose price has fallen will decrease, but with an inferior good, the substitution effect will be greater than the income effect and so overall there will be an increase in the quantity consumed of the good whose price has fallen. I.E and S.E goes in opposite direction. If the product is a Giffen good, the income effect will be greater than the substitution effect and so a fall in price will lead to decrease in the quantity demanded. I.E outweighs S.E effect.

Income and Substitution effect

Substitution effect

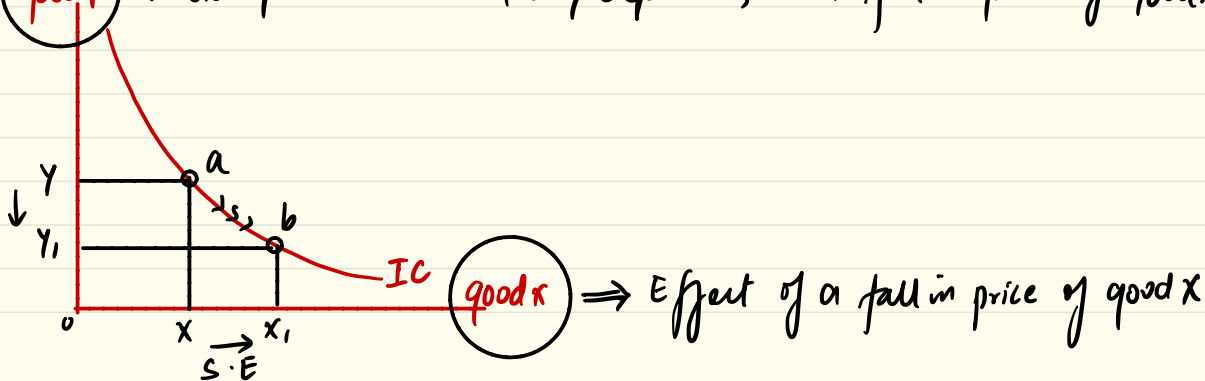
⇓
changing consumption bundles

⇓
From one combination to other

⇓
Movement along the indifference curve

⇓
due to relative change in price

good Y ⇒ Good Y becomes relatively expensive, due to fall in price of good X

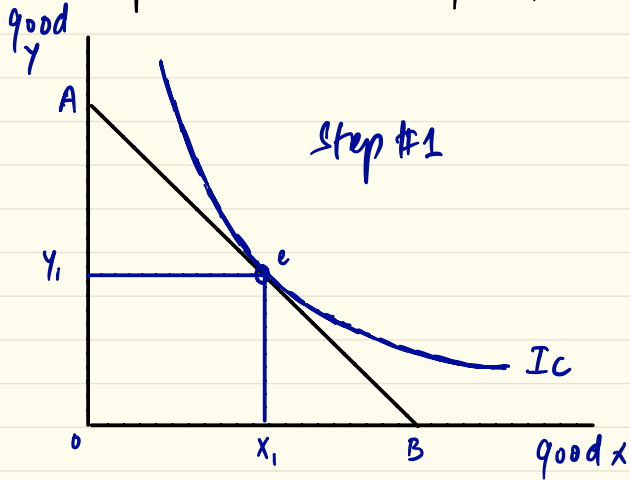


good X ⇒ Effect of a fall in price of good X

Case #1
Income and Substitution effect (Effect of a fall in price, in case of normal goods)

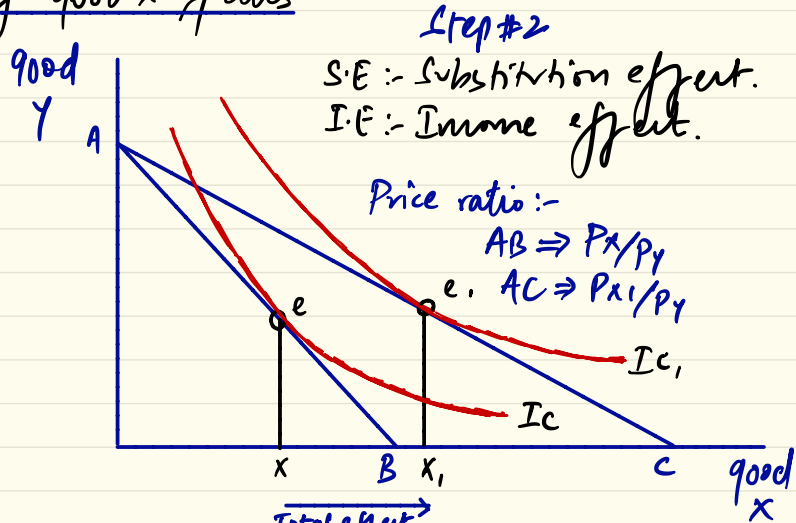
* This concept exists when asked for relative change in prices!

* Step #1 :- For e.g. price of good X falls



Initial Equilibrium
 $\Downarrow$

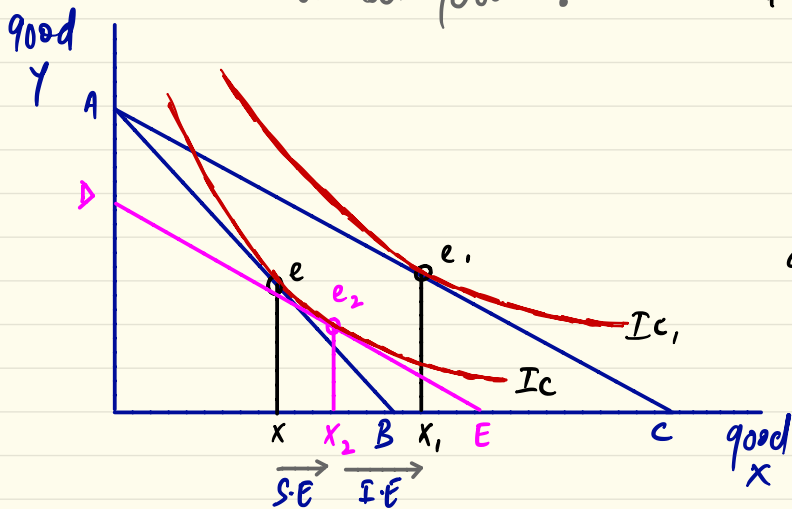
Rational consumer!



* This change shown from $X - X_1$ is total effect

$X - X_1$:- comprises of S.E and I.E

Normal goods !



* After total effect is shown, relative effect needs to be identified, which is done by drawing a parallel line to new budget line, just being tangent to old indifference curve!

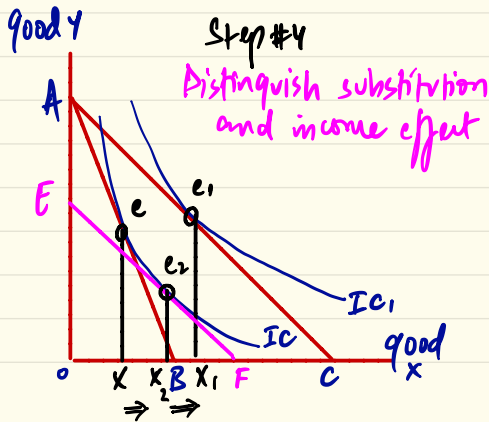
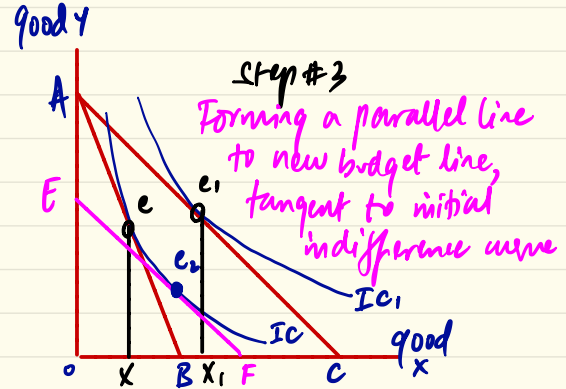
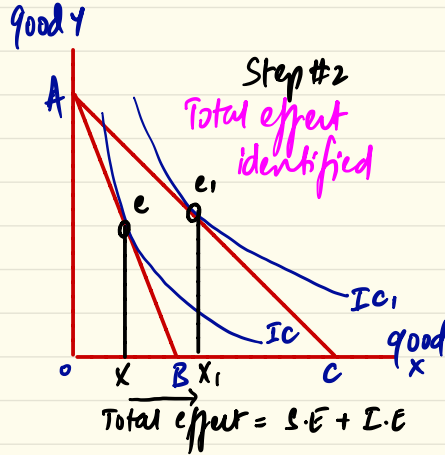
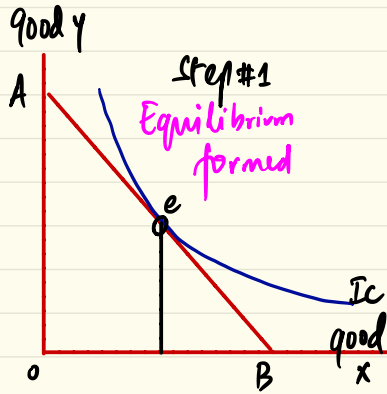
* Substitution effect represents relative change i.e. movement along the indifference curve i.e. ' e ' to ' e_2 '

* While income effect is from ' e_2 ' to ' e_1 '

* Income and substitution effect goes into same direction in case of normal goods.

lets break it down

Effect of a fall in price, in case of normal goods



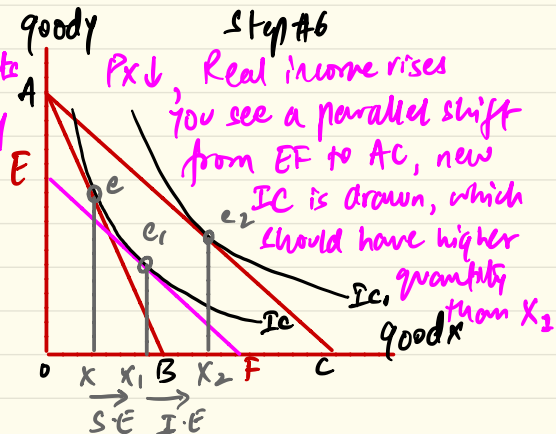
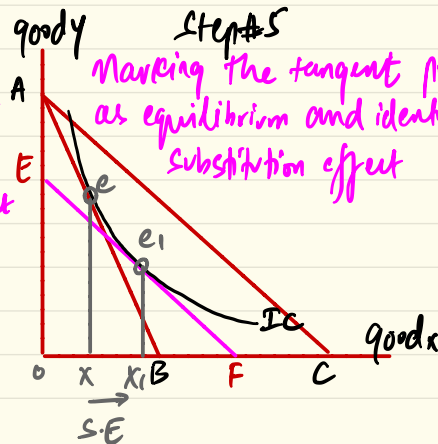
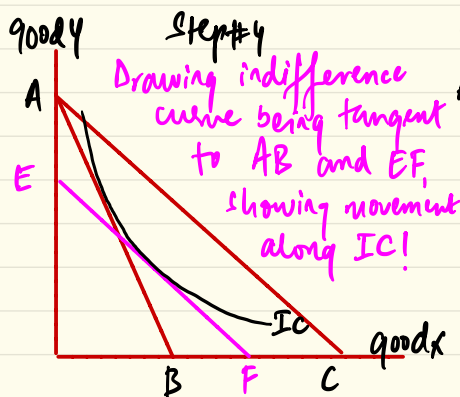
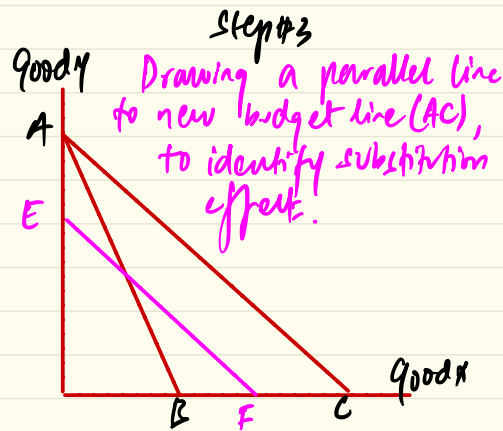
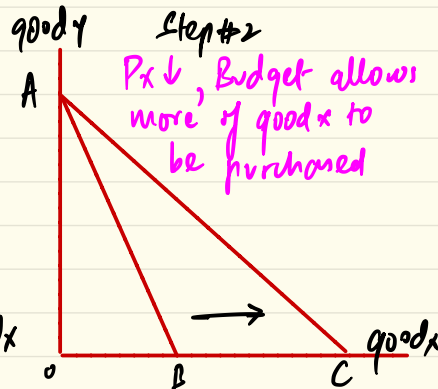
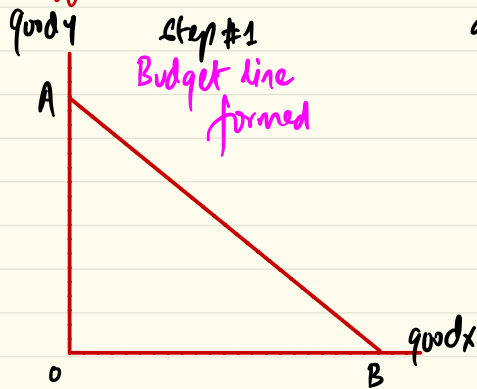
In case of normal goods, income and substitution effect goes in same direction

Substitution effect, it will be movement along the indifference curve, initial budget line and parallel budget line formed

Income effect will be a shift from parallel budget line formed to new budget line.

Another way to draw income and substitution effect (I prefer this way, but you must have a grip over previous slides)

Effect of a fall in price, in case of normal good



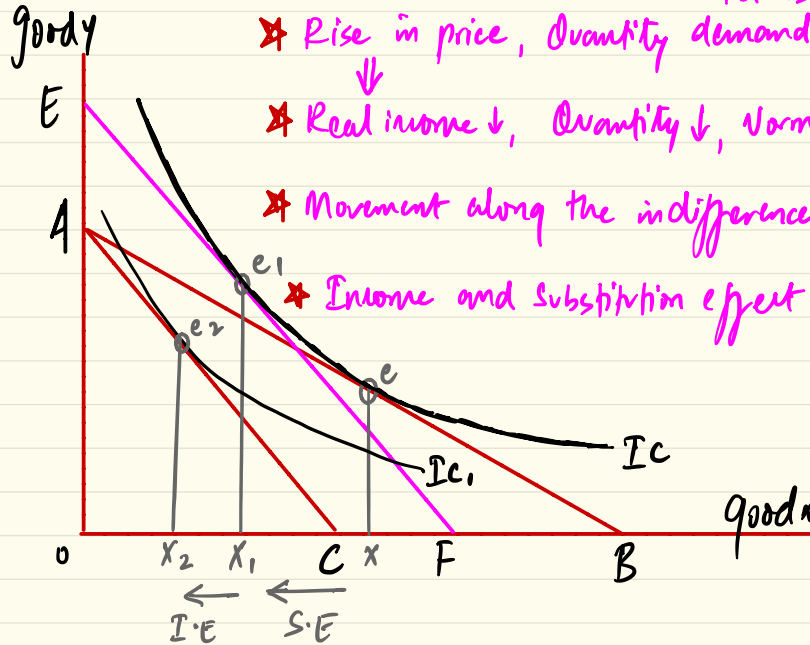
Case #2

Effect of a rise in price, in case of normal goods

Income and substitution effect

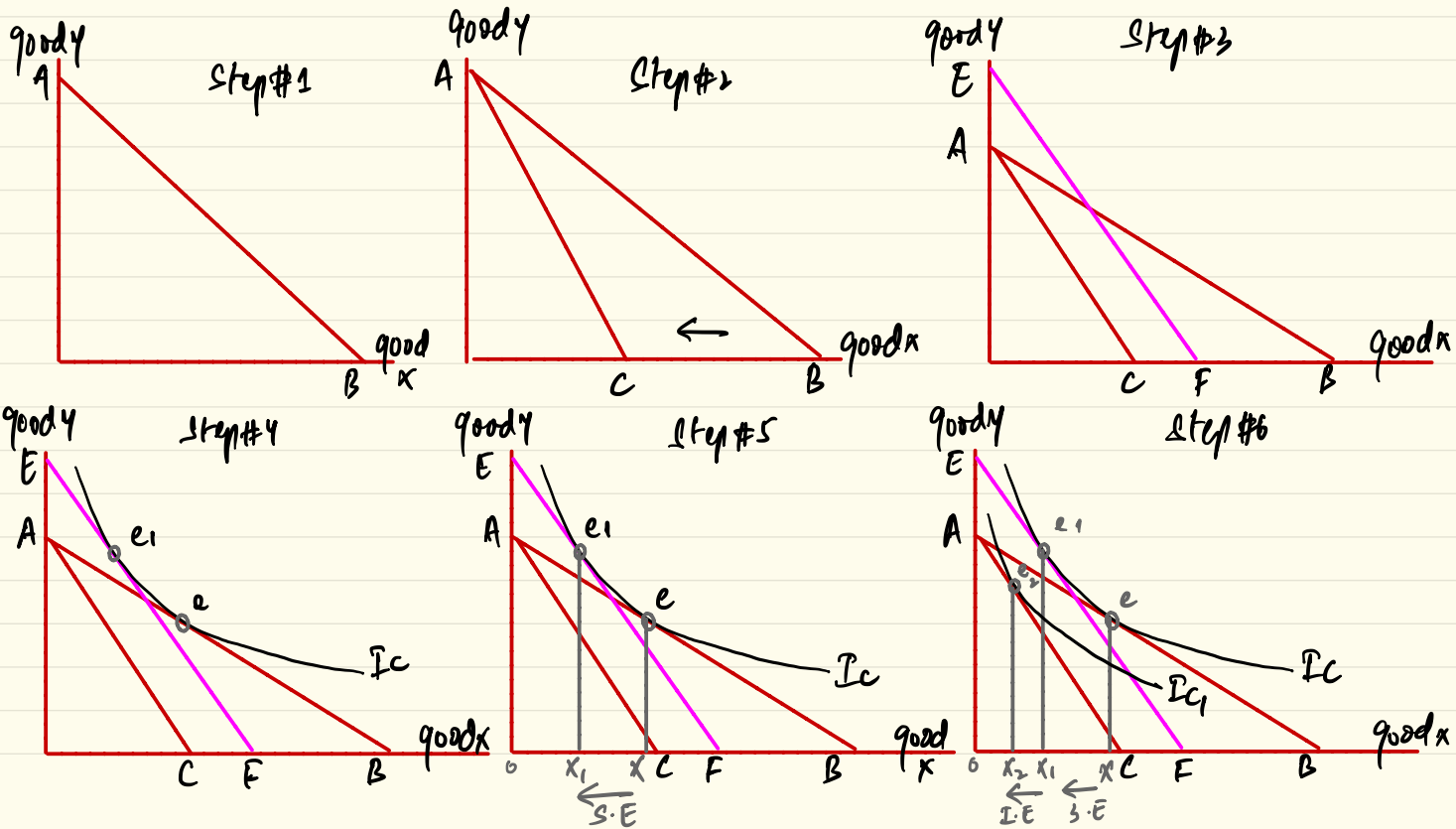
Points:-

- ★ Rise in price, Quantity demand reduce, Law of demand
- ★ Real income ↓, Quantity ↓, Normal goods, positive relationship
- ★ Movement along the indifference curve, Substitution effect
- ★ Income and substitution effect goes in same direction



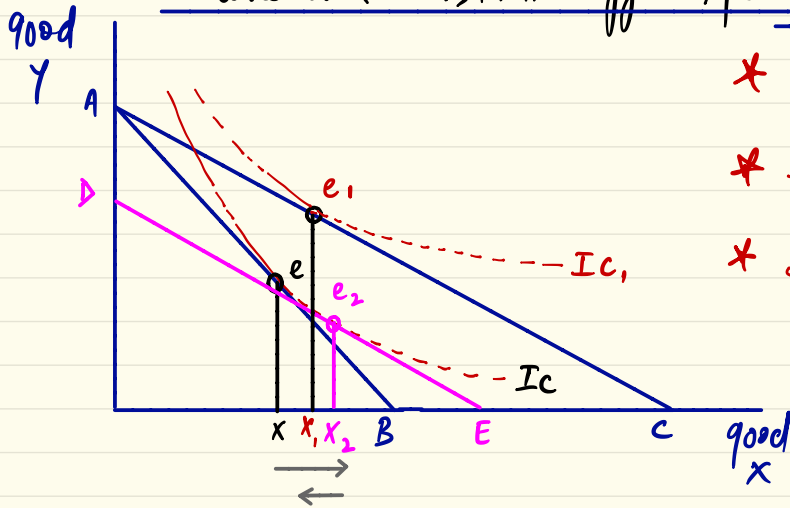
lets Break it down (Effect of a rise in price, in case of normal goods)

Income and Substitution effect.



Case #3

Income and Substitution effect for inferior good.



* Total effect :- $x - x_1$

* Substitution effect :- $x - x_2$

* Income effect :- $x_2 - x_1$

Effect of a fall in price!

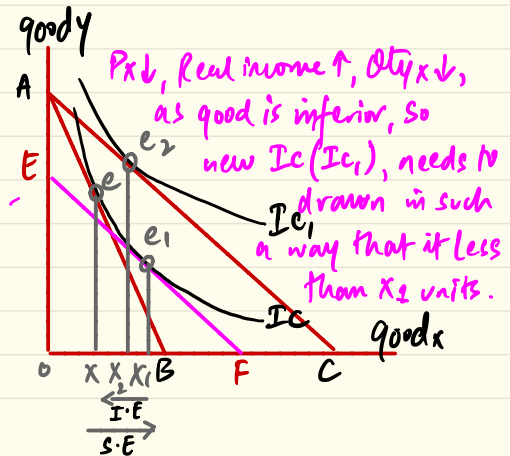
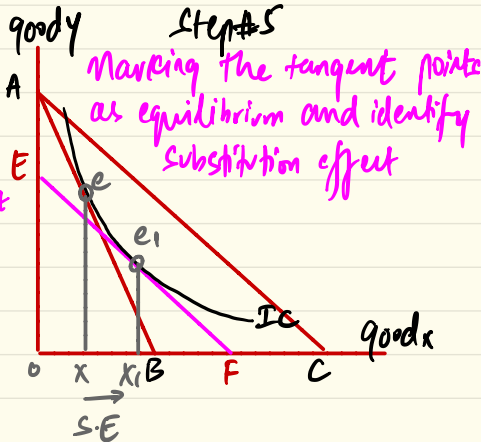
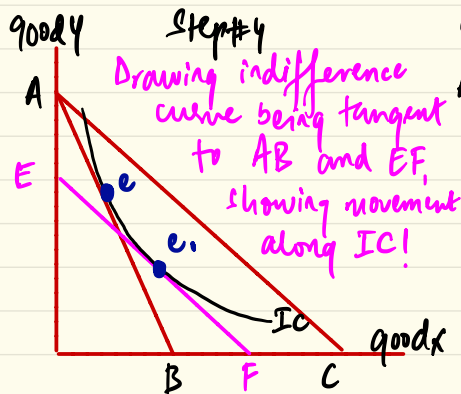
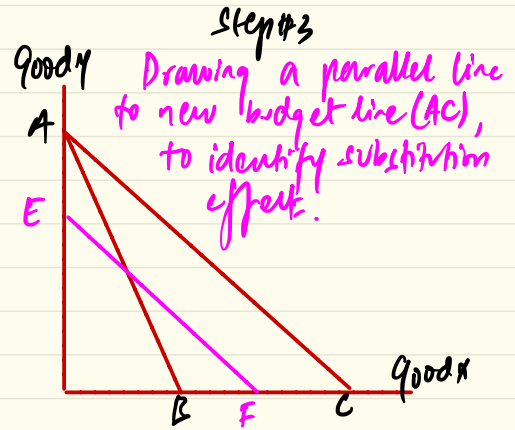
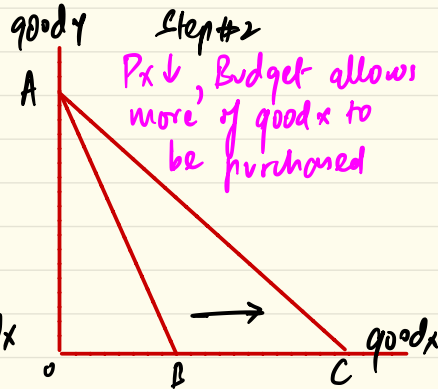
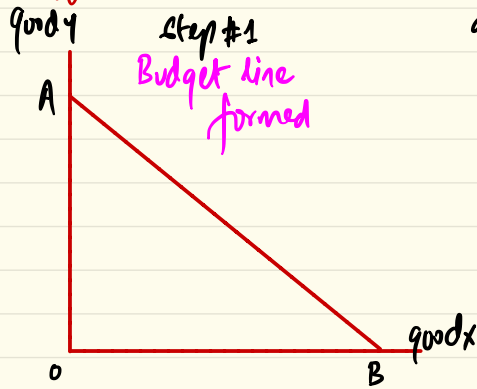
* In case of inferior goods, income and substitution effect goes in opposite direction!

$P_x \downarrow \Rightarrow QD_x \uparrow$ $\Rightarrow$ Law of demand (Substitution effect)

$\downarrow$
Real income $\uparrow \Rightarrow QD_x \downarrow$, as good is inferior

Another way to draw income and substitution effect

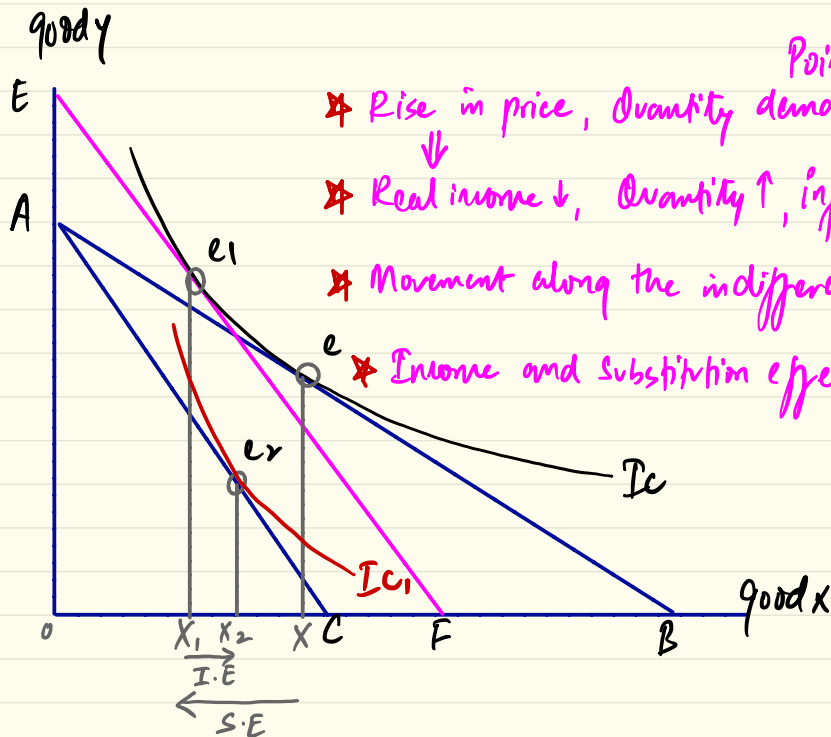
Effect of a fall in price, in case of inferior good



Case #4

Effect of a rise in price of a good, in case of inferior good

Income and Substitution effect.

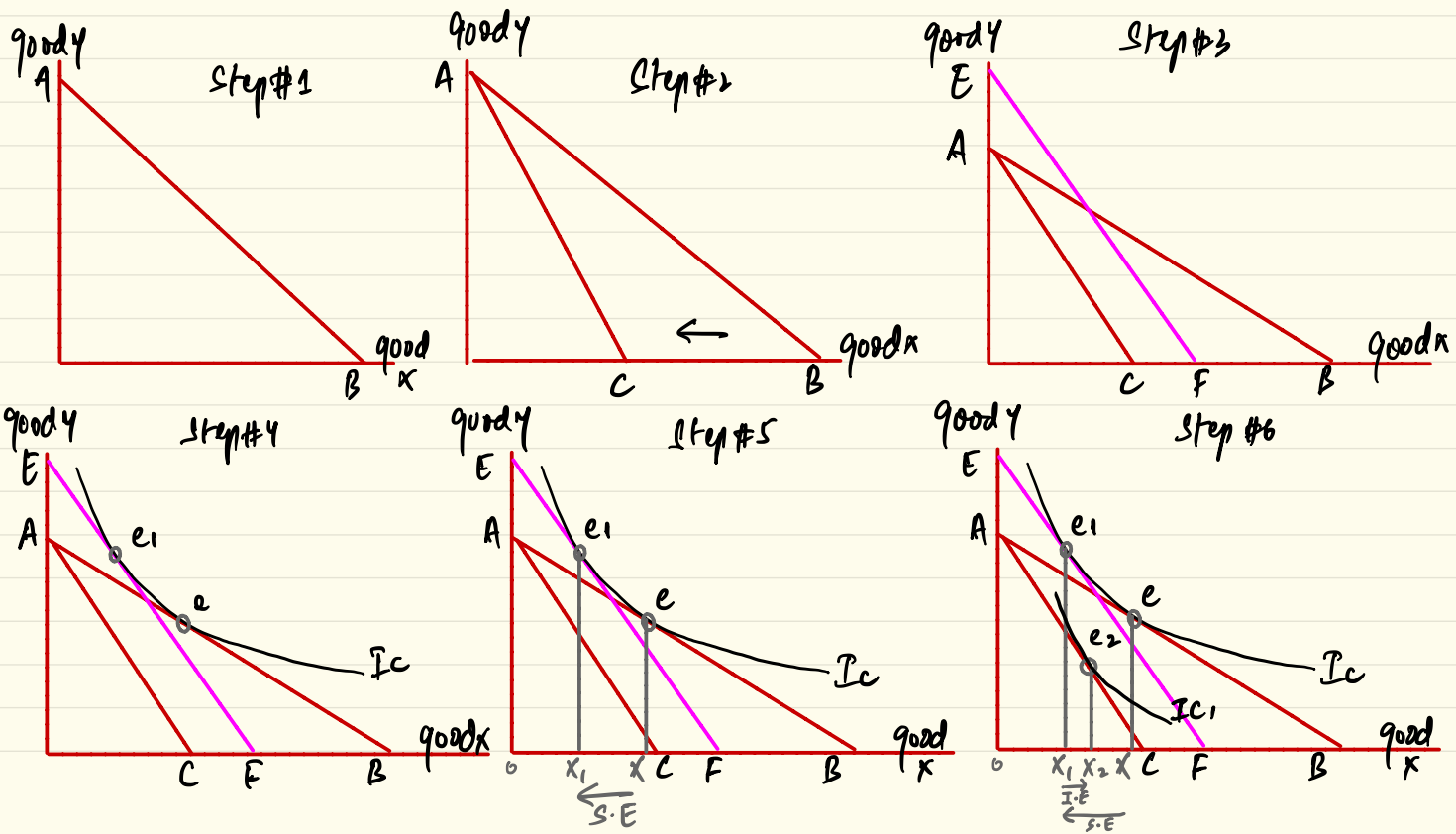


Points :-

- ✱ Rise in price, Quantity demand reduce, Law of demand
↓
- ✱ Real income ↓, Quantity ↑, inferior goods, Negative relationship
- ✱ Movement along the indifference curve, Substitution effect
- ✱ Income and Substitution effect goes in opposite direction

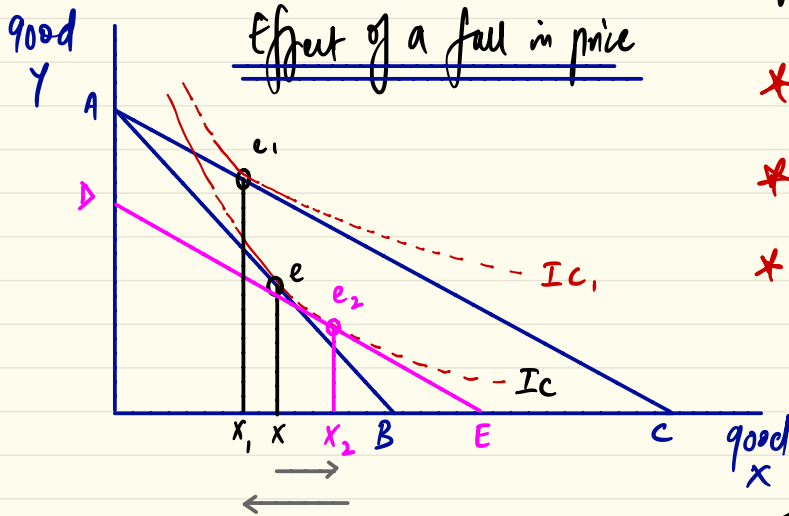
Let's Break it down (Effect of rise in price, in case of inferior good)

Income and Substitution effect



Case #5

Income and Substitution effect, Giffen good.



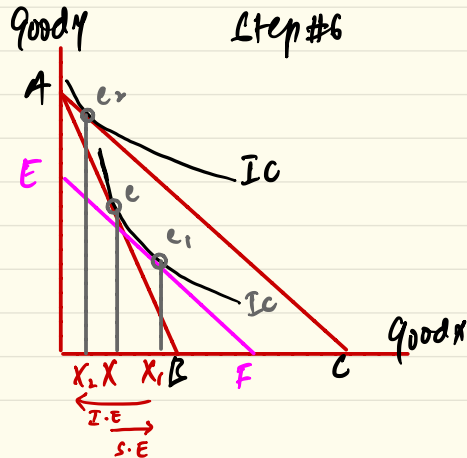
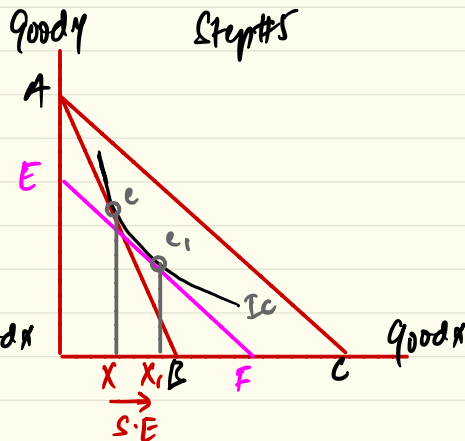
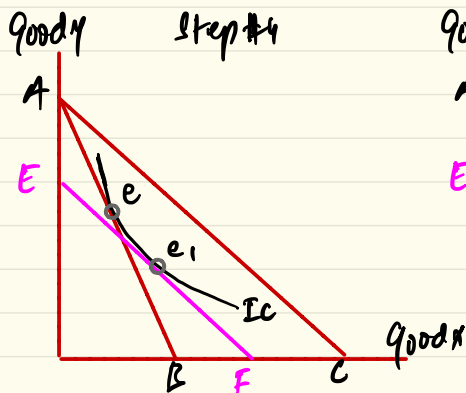
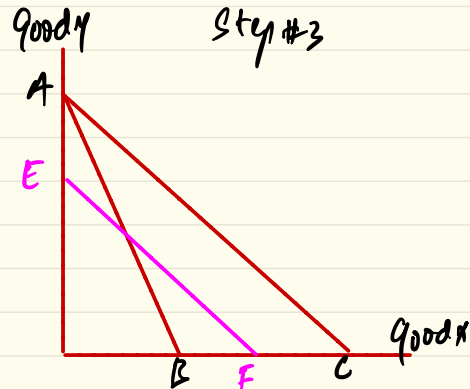
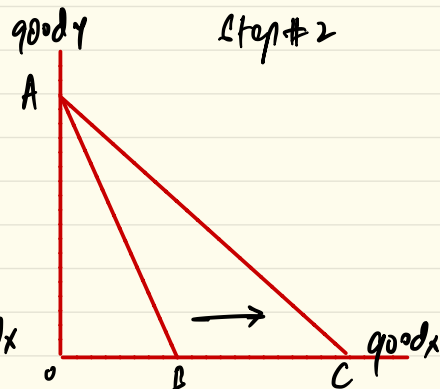
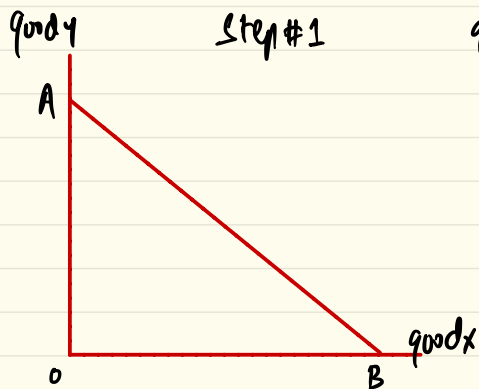
* Total effect :- $x - x_2$

* Substitution effect :- $x - x_1$

* Income effect :- $x_2 - x_1$

* When income effect outweighs substitution effect, it is represented the case of Giffen goods.

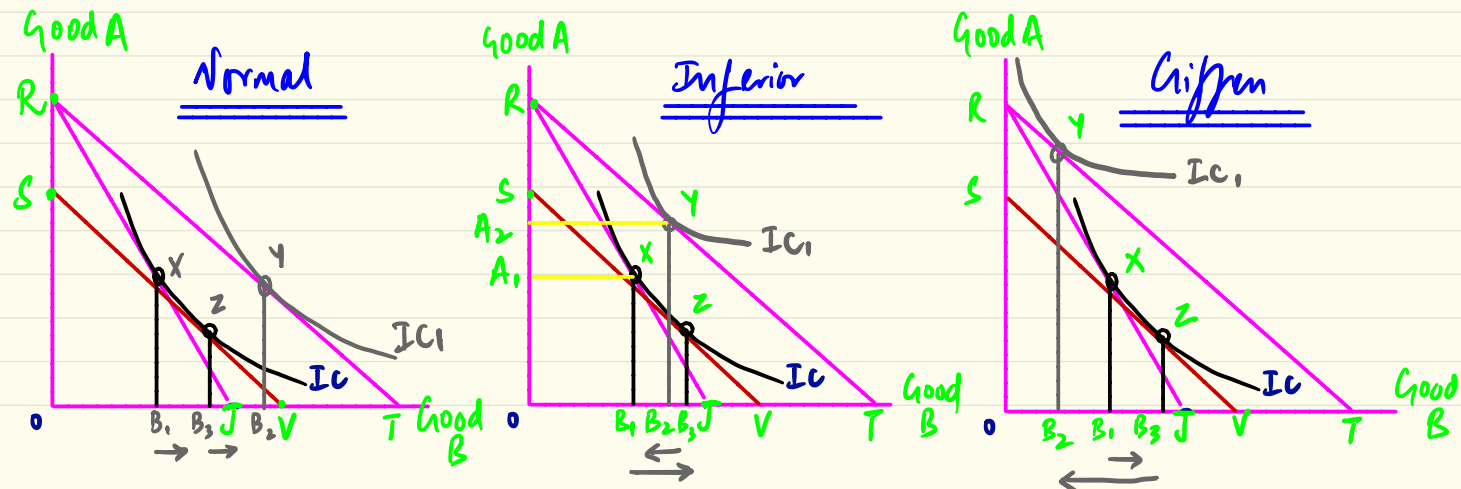
Effect of a fall in price, in case of giffon good (Income and Substitution effect)



Effect of a fall in price, income and substitution effect

It is possible to illustrate the S.E and I.E of the fall in the price of a good in a diagram. As described above, the effect of a fall in the price will, in each case, move the consumer from point X (consuming A_1, B_1) to point Y (A_2, B_2).

We begin by identifying the substitution effect. In order to do this we need to remove the income effect which is achieved by drawing in a new budget line, SV parallel to the new budget line (RT), but tangential to the original indifference curve I_1 . This leaves the individual on the original IC and shows how much of B would have been purchased had their real income not been increased. The movement from X to Z represents the S.E. The rest of the movement from X to Y is, therefore, the income effect.



It can be seen, therefore, that when the price of B is reduced, the change in quantity demanded, resulting from the S.E., is the movement from B_1 to B_3 and I.E from B_3 to B_2 .

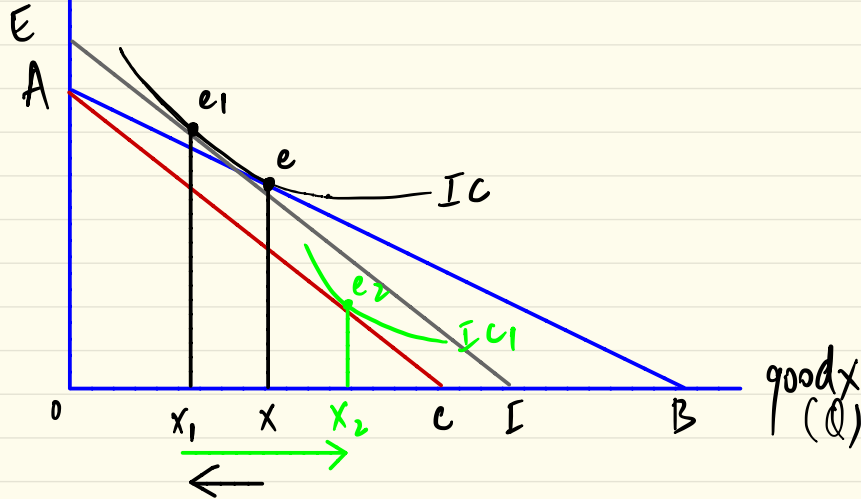
In case of normal good I.E and S.E goes in same direction, to bring an increase in quantity of B demanded. An inferior good is one for which the quantity demanded will fall as consumer's income rises. In this case S.E and I.E moves in opposite direction. The S.E increases the quantity demanded from B_1 to B_3 , but negative I.E moderates the increase to some extent and will reduce the consumption of B from B_3 to B_2 . Notice however, that even though S.E and I.E move in opposite directions, the final effect of the price reduction of good B is to increase the quantity of it which is demanded.

The final case is of Giffen good, regarded as goods with low quality which are important elements in the expenditure of those on low incomes. A good e.g. is a basic food such as rice, which forms a significant part of the diet of the poor in many countries. The argument, not accepted by all economists, is that when the price of rice falls sufficiently individual's real income will rise to an extent that they will be able to afford more attractive substitutes such as fresh fruit or vegetables to make up their diet and as a result they will actually purchase less rice even though its price has fallen. In this case, the negative income effect completely outweighs the substitution effect ($B_1 B_3$) ($B_3 B_2$)

Rise in price of good X (Giffen good) (Income and substitution effect)

good Y (Q)

* Income effect outweighs substitution effect

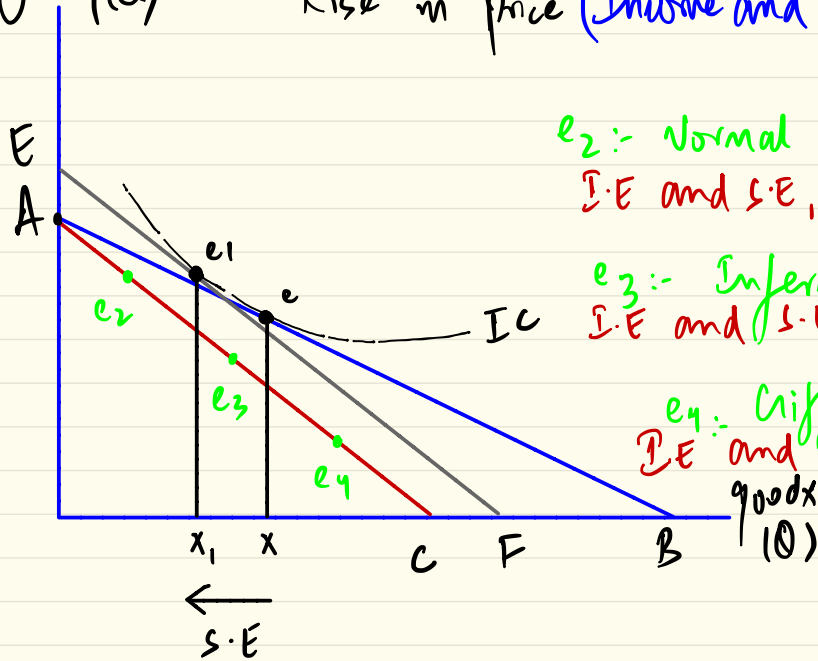


$x - x_1$: Substitution effect

$x_1 - x_2$: Income effect

good y (Q)

Rise in price (Income and substitution effect)



e_2 :- Normal good

I.E and S.E, same direction

e_3 :- Inferior good

I.E and S.E, opposite direction

e_4 :- Giffen good

I.E and S.E, outweighs

Limitations of the model of indifference curve

Although indifference curve analysis gets around the problem of measuring utility, when analysing consumer behaviour it has certain limitations.

To begin with, there is the issue of whether individuals can realistically be expected to be able to identify imaginary combinations of two goods which will give equal amounts of satisfaction and then compare them with other combination of same two goods which gives more or less satisfaction, which is what necessary to create an indifference map.

It is also the case that indifference curves are drawn on the basis of an individual's perception of the amount of utility that will be obtained from commodities, which might differ significantly from the satisfaction that is obtained when they are actually consumed.

Moreover, indifference curve analysis is unable to explain purchases of goods such as cars or computers or other consumer durables which are purchased infrequently with no consistent trade off against another product. Such goods may also only be purchased once. Finally, consumers may not act rationally when planning their expenditure.

Questions

Marks:- 24

Q1 Using consumer equilibrium, explain income and substitution effect of a fall in price of good, considering normal, inferior and giffen? 20

Ans:- Intro (I.E and S.E) - paragraphs given

Draw 3 diagrams

→ paragraphs given

Explain S.E once, make 3 paragraphs for income effect for different goods + limitations

Q2 Using consumer equilibrium, explain income and substitution effect of a rise in price of good, considering normal, inferior and giffen? 20

Ans:- Draw 3 diagrams (No Intro) → already given in Q1

Single paragraph for S.E

Separate paragraphs for I.E